

AI-Powered Employee Attrition Prediction System

Exploratory Data Analysis (EDA) Report

1. Abstract

Employee attrition is a major challenge for organizations as it leads to increased recruitment costs, training expenses, and productivity loss. This project analyzes employee data to identify factors contributing to attrition and provides insights that can help Human Resources (HR) departments improve employee retention strategies.

2. Problem Statement

Organizations often struggle to predict which employees are likely to leave. The objective of this analysis is to:

- Determine the overall attrition rate.
 - Identify departments with high employee turnover.
 - Analyze the impact of salary, experience, age, and job satisfaction on attrition.
 - Discover key factors influencing employee retention.
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3. Dataset Description

Dataset: IBM HR Analytics Employee Attrition & Performance

- Total Records: 1470 Employees
- Total Features: 35
- Target Variable: Attrition (Yes/No)

Key Features:

- Age
 - Gender
 - Department
 - Monthly Income
 - Job Satisfaction
 - Total Working Years
 - Years at Company
 - Overtime
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4. Data Cleaning and Preprocessing

The dataset was examined for data quality issues.

Findings:

- Missing Values: 0
- Duplicate Records: 0
- Data Types Verified
- Categorical Features Encoded for Machine Learning

The dataset was found to be clean and ready for analysis.

5. Overall Attrition Analysis

Employee Attrition Distribution:

Attrition Status	Count
No	1233
Yes	237

Attrition Rate:

- Employees Stayed: 83.87%
- Employees Left: 16.12%

Observation:

The dataset is moderately imbalanced, with approximately 16% of employees leaving the organization.

6. Department-wise Attrition

Department	Attrition Rate	Yes	No
Sales	Highest	92	354
Human Resources	Moderate	12	51
Research & Development	Lowest	133	828

Observation:

The Sales department experiences the highest employee turnover, indicating possible issues related to workload, targets, or job satisfaction.

7. Gender-wise Attrition

Gender	Attrition Rate	Yes	No
Male	Slightly Higher	150	732
Female	Slightly Lower	87	501

Observation:

Gender does not appear to be a major factor influencing employee attrition.

8. Salary Impact on Attrition

Average Monthly Income:

Attrition Status	Average Monthly Income
Stayed	\$6,832.73
Left	\$4,787.09

Observation:

Employees with lower salaries are more likely to leave the organization.

9. Experience Impact on Attrition

Average Total Working Years:

Attrition Status	Average Experience
Stayed	12 Years
Left	8 Years

Observation:

Employees with less work experience are more likely to resign.

10. Age Analysis

Average Age:

Attrition Status	Average Age
Stayed	38 Years
Left	33 Years

Observation:

Younger employees exhibit higher attrition rates compared to older employees.

11. Job Satisfaction Analysis

Average Job Satisfaction Score:

Attrition Status	Satisfaction Score
Stayed	2.7
Left	2.4

Observation:

Low job satisfaction is strongly associated with employee attrition.

12. Correlation Analysis

A correlation heatmap was generated to examine relationships among numerical variables.

Key Findings:

- Monthly Income positively correlates with Total Working Years.
 - Years at Company positively correlates with Age.
 - Job Satisfaction negatively correlates with Attrition.
 - Overtime positively correlates with Attrition.
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13. Key Insights

1. Overall attrition rate is approximately 16%.
 2. Sales department experiences the highest employee turnover.
 3. Lower salaries significantly increase attrition risk.
 4. Employees with lower job satisfaction are more likely to leave.
 5. Younger and less experienced employees show higher attrition rates.
 6. Overtime contributes significantly to employee turnover.
 7. Research & Development employees demonstrate the highest retention.
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14. Business Recommendations

- Improve compensation for low-income employees.
 - Increase employee engagement and job satisfaction programs.
 - Reduce excessive overtime workload.
 - Focus retention efforts on younger employees.
 - Provide career growth opportunities for employees in high-risk departments.
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15. Conclusion

The EDA revealed several factors that influence employee attrition, including salary, experience, age, job satisfaction, and overtime. These insights will be utilized in the Machine Learning phase to build predictive models capable of identifying employees at risk of leaving the organization.